

Hands-on Lab: COUNT, DISTINCT, LIMIT



Estimated time needed: 30 minutes

In this lab, you will learn a few useful expressions that are used with SELECT statements. First, you will learn COUNT, which is an aggregate function that retrieves the number of rows that matches the query criteria. Next, you will learn DISTINCT, which is used to remove duplicate values from a specified result set and only return the unique values. Lastly, you will learn LIMIT, which is used for restricting the number of rows retrieved from the table.

Software used in this lab

In this lab, you will use [Datasette](#), an open-source multi-tool for exploring and publishing data.

Database used in this lab

The database used in this lab comes from the following data set source: [Film Locations in San Francisco](#) under a [PDDL: Public Domain Dedication and License](#).

Objectives

After completing this lab, you will be able to:

- Retrieve the number of rows that match a query criteria
- Remove duplicate values from a result set and return the unique values
- Restrict the number of rows retrieved from a table

Exploring the Database

Let us first explore the **SanFranciscoFilmLocations** database using the **Datasette** tool:

1. If the first statement listed below is not already in the Datasette textbox on the right, then copy the code below by clicking the little copy button on the bottom right of the code block and then paste it into the textbox of the Datasette tool using either **Ctrl+V** or right-click in the text box and choose **Paste**.

```
SELECT * FROM FilmLocations;
```

home / Practice SQL / SanFranciscoFilmLocations

Practice SQL

Database: SanFranciscoFilmLocations

```
1 SELECT * FROM FilmLocations;
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

2. Click **Submit Query**.
3. Now, you can scroll down the table and explore all the columns and rows of the **FilmLocations** table to get an overall idea of the table.

Title	ReleaseYear	Locations	FunFacts	ProductionCompany	Distributor	Director	Writer	Actor1	Actor2	Actor3
180	2011	Epic Roasthouse (399 Embarcadero)		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Mason & California Streets (Nob Hill)		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Justin Herman Plaza		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	200 block Market Street		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	City Hall		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Polk & Larkin Streets		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Randall Museum		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	555 Market St.		SPI Cinemas		Jayendra	Umarji	Siddarth	Nithya	Priya

Support

4. These are the column attribute descriptions from the **FilmLocations** table:

```

FilmLocations(
  Title:           titles of the films,
  ReleaseYear:     time of public release of the films,
  Locations:       locations of San Francisco where the films were shot,
  FunFacts:        funny facts about the filming locations,
  ProductionCompany: companies who produced the films,
  Distributor:     companies who distributed the films,
  Director:        people who directed the films,
  Writer:          people who wrote the films,
  Actor1:          person 1 who acted in the films,
  Actor2:          person 2 who acted in the films,
  Actor3:          person 3 who acted in the films
)

```

Using COUNT statement

Let us go through some examples of COUNT-related queries.

1. Suppose we want to count the number of records or rows of the "FilmLocations" table. The query for this would be:

```
SELECT COUNT(*) FROM FilmLocations;
```

Copy the code above and paste it to the query box of the Datasette tool. Then click **Submit query**. Your output result set should look like the image below:

home / Practice SQL / SanFranciscoFilmLocations

Practice SQL

Database: SanFranciscoFilmLocations

1 SELECT COUNT(*) FROM FilmLocations;

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

Support

SELECT COUNT(*) FROM FilmLocations

COUNT(*)

3414

Powered by [Datasette](#)

2. We want to count the number of locations of the films. But we also want to restrict the output result set so that we only retrieve the number of locations of the films written by a certain writer. The query for this can be written as:

```
SELECT COUNT(Locations) FROM FilmLocations WHERE Writer="James Cameron";
```

Copy the code above and paste it to the textbox of the Datasette tool. Then click **Submit query**.

home / Practice SQL / SanFranciscoFilmLocations

Practice SQL

Database: SanFranciscoFilmLocations

```
1 SELECT COUNT(Locations) FROM FilmLocations WHERE Writer="James Cameron";
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

SELECT COUNT(Locations) FROM FilmLocations WHERE Writer="James Cameron"

COUNT(Locations)
48

Powered by [Datasette](#)

Using DISTINCT statement

In this exercise, you will go through some examples of using DISTINCT in queries.

1. Assume that we want to retrieve the titles of all films in the table so that duplicates will be discarded in the output result set.

```
SELECT DISTINCT Title FROM FilmLocations;
```

Copy the code above and paste it to the textbox of the Datasette tool. Then click **Submit query**.

Your output resultset should look like the image below:

Practice SQL

Database: SanFranciscoFilmLocations

```
1 SELECT DISTINCT Title FROM FilmLocations;
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

SELECT DISTINCT Title FROM FilmLocations

Title

180
24 Hours on Craigslist
A Night Full of Rain
About a Boy
Age of Adaline
After the Thin Man
Ant-Man
Americana
Another 48 Hours
Around the Fire
Attack of the Killer Tomatoes

Support

2. We want to retrieve the count of release years of the films produced by a specific company so that duplicate release years of those films will be discarded in the count.

```
SELECT COUNT(DISTINCT ReleaseYear) FROM FilmLocations WHERE ProductionCompany="Warner Bros. Pictures";
```

Copy the code above and paste it to the textbox of the Datasette tool. Then click **Submit query**.

Your output resultset should look like the image below:

The screenshot shows the Datasette SQL practice tool interface. At the top, there is a breadcrumb navigation: "home / Practice SQL / SanFranciscoFilmLocations". Below this is the title "Practice SQL" and the database name "Database: SanFranciscoFilmLocations". A text area contains the SQL query: `1 SELECT COUNT(DISTINCT ReleaseYear) FROM FilmLocations WHERE ProductionCompany="Warner Bros. Pictures";`. Below the query is a tip: "Tip: Autocomplete with Ctrl+Enter or Cmd+Enter". A "Submit query" button is located below the tip. The "Results" section shows a green message: "All commands ran successfully". Below this, the query is repeated: `SELECT COUNT(DISTINCT ReleaseYear) FROM FilmLocations WHERE ProductionCompany="Warner Bros. Pictures"`. The result is displayed as a table with one row:

COUNT(DISTINCT ReleaseYear)
14

. A "Support" button is visible on the right side of the results section. At the bottom, it says "Powered by Datasette".

Using LIMIT statement

In this exercise, you will first go through some examples of using LIMIT in queries.

1. Retrieve only the first 25 rows from the table so that rows other than those are not in the output result set.

```
SELECT * FROM FilmLocations LIMIT 25;
```

Copy the code above and paste it to the textbox of the Datasette tool. Then click **Submit query**.

Your output resultset should look like the image below:

Practice SQL

Database: SanFranciscoFilmLocations

```
1 SELECT * FROM FilmLocations LIMIT 25;
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

Support

```
SELECT * FROM FilmLocations LIMIT 25
```

Title	ReleaseYear	Locations	FunFacts	ProductionCompany	Distributor	Director	Writer	Actor1	Actor2	Actor3
180	2011	Epic Roasthouse (399 Embarcadero)		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Mason & California Streets (Nob Hill)		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand
180	2011	Justin Herman Plaza		SPI Cinemas		Jayendra	Umarji Anuradha, Jayendra, Aarthi Sriram, & Suba	Siddarth	Nithya Menon	Priya Anand

2. Now, we want to retrieve 15 rows from the table starting from row 11.

```
SELECT * FROM FilmLocations LIMIT 15 OFFSET 10;
```

Copy the code above and paste it to the textbox of the Datasette tool. Then click **Submit query**.

Your output resultset should look like the image below:

Practice SQL

Database: SanFranciscoFilmLocations

```
1 SELECT * FROM FilmLocations LIMIT 15 OFFSET 10;
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

Support

```
SELECT * FROM FilmLocations LIMIT 15 OFFSET 10
```

Title	ReleaseYear	Locations	FunFacts	ProductionCompany	Distributor	Director	Writer	Actor1	Actor2	Actor3
A Night Full of Rain	1978	Fairmont Hotel (950 Mason Street, Nob Hill)	In 1945 the Fairmont hosted the United Nations Conference on International Organization as delegates arrived to draft a charter for the organization. The U.S. Secretary of	Liberty Film	Warner Bros. Pictures	Lina Wertmuller	Lina Wertmuller	Candice Bergen	Giancarlo Gianni	

Practice exercises

COUNT

1. Retrieve the number of locations of the films which are directed by Woody Allen.

- Hint
- Query Solution
- Output

2. Retrieve the number of films shot at Russian Hill.

- ▶ Hint
- ▶ Query Solution
- ▶ Output

3. Retrieve the number of rows having a release year older than 1950 from the "FilmLocations" table.

- ▶ Hint
- ▶ Query Solution
- ▶ Output

Practice exercises

DISTINCT

1. Retrieve the names of all unique films released in the 21st century and onwards, along with their release years.

- ▶ Hint
- ▶ Query Solution
- ▶ Output

2. Retrieve the directors' names and their distinct films shot at City Hall.

- ▶ Hint
- ▶ Query Solution
- ▶ Output

3. Retrieve the number of distributors who distributed films with the 1st actor, Clint Eastwood.

- ▶ Hint
- ▶ Query Solution
- ▶ Output

Practice exercises

LIMIT

1. Retrieve the names of the first 50 films.

- ▶ Hint
- ▶ Query Solution
- ▶ Output

2. Retrieve the first 10 film names released in 2015.

- ▶ Hint
- ▶ Query Solution
- ▶ Output

3. Retrieve the next 3 film names that follow after the first 5 films released in 2015.

- ▶ Hint
- ▶ Query Solution
- ▶ Output

Conclusion

Congratulations! You have completed this lab.

You are now able to:

- Use COUNT statements to determine the number of entries in a database based on filtering conditions.
- Use DISTINCT statements to determine the unique entries in a database based on filtering conditions.
- Use LIMIT statements to restrict the response to a desired set of rows based on filtering conditions.
- Use a combination of these statements to execute more complex queries on the database.

Author(s)

[Sandip Saha Joy](#)

Other Contributor(s)

[Abhishek Gagneja](#)

© IBM Corporation 2023. All rights reserved.